Hung **Pham**

AI for Science · Generative Modeling • Quantum Chemistry • GPU-Accelerated Simulation

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Personal statement

I build intelligent systems for scientific discovery, combining AI, high-performance computing, and quantum chemistry. My experience spans developing new scientific methods, building GPU-accelerated simulation software, training generative AI models for chemistry, and designing agentic R&D systems. I enjoy solving problems that require both deep scientific understanding and large-scale software engineering.

Experiences

Research Scientist – ByteDance Seed, AI for Science (San Jose, CA)

Mar 2022 - Current

Focus: Generative modeling for chemistry • Quantum embedding • GPU-accelerated quantum chemistry

- Lead **Trèfle** – flow matching and reinforcement learning for reaction mechanisms: a one-step model that surpasses state-of-the-art accuracy on transition-state and reaction-outcome prediction at **30**× the throughput of multi-step baselines (arXiv:2609.32073).
- Built and led **ByteCRN** end to end – a reaction-network exploration platform pairing a conventional DFT workflow with a generative counterpart that proposes and validates transition states, giving **10–100**× acceleration and pruning 70–90% of enumerated reactions without loss of fidelity. Wrote most of the initial codebase; now lead a team of Ph.D. researchers on it, owning architecture, code review, and engineering standards (arXiv:2606.21002).
- Lead developer of **FEMION**, a GPU-accelerated *ab initio* quantum embedding framework extending the method to metallic (gapless) systems – reached **576-atom** copper supercells with accuracy matching diffusion Monte Carlo on Cu(111) surface chemistry. Corresponding author (arXiv:2508.13036).
- Core contributor to **ByteQC**, the lab's open-source GPU-accelerated quantum chemistry package – built the GPU random-phase approximation (RPA) module with MPI parallelization over k-points (*WIREs Comput. Mol. Sci.* 2025).
- Technical lead on several projects, setting direction and reviewing the work of other scientists and engineers; mentor Ph.D. interns.

Founding Advisor – VBrain (Hanoi, Vietnam)

Sep 2025 - Current

Focus: Agentic AI systems • Multi-agent collaboration • R&D operating system

- Set technical direction for an **agentic R&D operating system**: system architecture, technology decisions, and technical hiring.
- Contribute to company building alongside the technical work – **top 10** at Vietnam AI Stars 2026 and selected for the VietLeap AI Accelerator 2025, run by Vietnam's National Innovation Center (NIC).

Postdoctoral Research Scientist – Columbia University (New York, NY)

Jun 2021 - Mar 2022

Advisor: David Reichman | *Focus*: Auxiliary-field quantum Monte Carlo • Quantum embedding for periodic solids

- Co-authored the field's benchmark assessment of twenty years of AFQMC in quantum chemistry – *J. Chem. Theory Comput.* **Front Cover, ACS Editors' Choice** (2022).

Research Assistant – University of Minnesota, Twin Cities (Minneapolis, MN)

Jan 2017 - May 2021

Advisor: Laura Gagliardi | *Focus*: Quantum embedding • Multireference electronic structure • Materials design

- Formulated the theory and wrote the entire codebase for **pDMET**, extending density matrix embedding theory to periodic solids; also authored the first DMET formulation with an MCSCF (CASSCF) solver – first author on both (*J. Chem. Theory Comput.* 2018, 2020).
- Built **MCU** and **pyWannier90**, open-source tools for band structure, topology, and Wannier analysis in periodic DFT.
- Mentored a Ph.D. student on a multireference quantum embedding study of excited states in crystalline point defects (*J. Phys. Chem. Lett.* 2021, co-first author).
- Predicted lead-free double perovskites for photovoltaics (*Nanoscale* 2019) and a topological insulating phase in pentagonal COFs (*Chem. Mater.* 2021, corresponding author).

Research Assistant – MANAR & ICST (Ho Chi Minh City, Vietnam)

Oct 2011 - Jul 2015

Advisor: Nguyen-Nguyen Pham-Tran | *Focus*: Periodic DFT • Grand canonical Monte Carlo • Reticular chemistry

- First author on a systematic DFT study of band-gap engineering in metal-organic frameworks through organic-linker functionalization (*J. Phys. Chem. C* 2014).
- Ran periodic DFT and grand canonical Monte Carlo simulations for gas-storage materials design, as **principal investigator** on a university research grant funding part of the work.
- Visiting researcher in Omar Yaghi's lab (2025 Nobel Laureate in Chemistry), UCLA – MOF synthesis and single-crystal XRD characterization.

Education and Training

Ph.D. in Quantum Chemistry – University of Minnesota, Twin Cities

Aug 2015 - May 2021

Dissertation: Electronic Structure Theories for Periodic Systems

B.Sc. in Chemistry – Vietnam National University Ho Chi Minh City - University of Science

Sep 2007 - Sep 2011

Honors Program

Selected Software

ByteCRN — lead developer, DFT + generative AI for reaction-network exploration *arXiv:2606.21002*
FEMION — lead developer, Quantum embedding for metals, GPU random phase approximation *arXiv:2508.13036*
ByteQC — contributor, GPU-accelerated quantum chemistry at scale. CUDA • MPI *128* ★
MCU — lead developer, Band structure, topology & Wannier analysis for periodic DFT *53* ★
pyWannier90 — lead developer, Wannier90 interface for VASP and PySCF *43* ★
pDMET — sole developer, Density matrix embedding theory for periodic systems (*J. Chem. Theory Comput.* 2020) *19* ★
PySCF — contributor, Core open-source quantum chemistry framework for Python *1.6k* ★
ipie — contributor, Python auxiliary-field quantum Monte Carlo package *69* ★

Technical Skills

How I build: Spec-driven & agentic development • I architect, specify, and review

Languages: Python | C/C++ | CUDA | Fortran | Bash

AI & Machine Learning: Flow Matching | Score-based & Diffusion Models | Generative Modeling | Reinforcement Learning | LLM APIs & Agentic Systems | MCP | RAG | Evals & Fine-tuning | PyTorch | GNNs

HPC & Infrastructure: GPU Computing | MPI | SLURM | Workflow Orchestration | High-Throughput Screening | Docker | Git | CI/CD

Quantum chemistry packages: VASP | Gaussian | CRYSTAL | Quantum Espresso | CP2K | Wannier90

Publications

Full list of publications can be found in my **Google Scholar (clickable)** profile | **Note:** * Corresponding author | †Co-first author

I. Generative Modeling & AI for Chemistry

1. Y. Li, Z. Sun, K. Yu, W. Yan, M. Gerstein, **H. Q. Pham***, **arXiv:2609.32073**, 2026.
2. Z. Sun, C. Hu, K. Lin, J. Li, C. Cao, D. Lv, J. Chen, W. Ren, **H. Q. Pham***, **arXiv:2606.21002**, 2026.

II. Parallel & GPU-Accelerated Software

1. Z. Guo, Z. Huang, Q. Chen, J. Shao, G. Liu, **H. Q. Pham**, Y. Huang, C. Cao, J. Chen, D. Lv, **WIREs Comput. Mol. Sci.**, 2025, 15.
2. Y. Huang, Z. Guo, **H. Q. Pham**, D. Lv, **arXiv:2406.08314**, 2024.

III. Quantum Embedding and Quantum Algorithms

1. C. Cao†, **H. Q. Pham***†, Z. Guo, Y. Zhang, Z. Huang, X. Wen, J. Chen, D. Lv, **arXiv:2508.13036**, 2025.
2. Z. Huang, Z. Guo, C. Cao, **H. Q. Pham**, X. Wen, G. H. Booth, J. Chen, D. Lv, **Nat. Commun.**, 2025, 16.
3. C. Cao, J. Sun, X. Yuan, H.-S. Hu, **H. Q. Pham***, D. Lv, **npj Comput. Mater.**, 2023, 78.
4. **H. Q. Pham**, M. R. Hermes, L. Gagliardi, **J. Chem. Theory Comput.**, 2020, 16, 130.
5. **H. Q. Pham**, V. Bernales, L. Gagliardi, **J. Chem. Theory Comput.**, 2018, 14, 1960.

IV. Quantum Monte Carlo Method Development

1. J. Lee, **H. Q. Pham**, D. R. Reichman, **J. Chem. Theory Comput.**, 2022, 18, 12, 7024 **(Front Cover, ACS Editors' Choice)**.
2. **H. Q. Pham***, R. Ouyang, and D. Lv, **J. Chem. Theory Comput.**, 2024, 20, 9, 3524.

V. Computational Design of Novel Materials

1. **H. Q. Pham***, N.-N. Pham-Tran, **Chem. Mater.**, 2021, 33, 4488.
2. **H. Q. Pham**, T. Mai, N.-N. Pham-Tran, Y. Kawazoe, H. Mizuseki, D. Nguyen-Manh, **J. Phys. Chem. C**, 2014, 118 (9), 4567.

Honors & Awards

VEF Fellowship, Vietnam Education Foundation *2015*
Grant (Principal Investigator): VNU C2013-50-03, Vietnam National University Ho Chi Minh City *2013*
Academic Excellence Scholarship, Viet Nam National University Ho Chi Minh City - University of Science *2008-2011*
Lawrence S. Ting Scholarship, Lawrence S. Ting Memorial Fund *2008*
3rd Place (Shared), Vietnam National Olympiad in Chemistry for High-School Students *2007*

Review Service

Peer reviewer: J. Chem. Theory Comput. | J. Chem. Phys. | Phys. Chem. Chem. Phys. | Chem. Phys. Lett. | IEEE Trans. Nanotechnology | Solid State Commun. | New J. Chem. | Digital Discovery | J. Open Source Softw. | Future Drug Discov.